

Assignment 3: Comparing Classification Models

Course: MBAI 5310G-001 AI Programming | Instructor: Zahra Atf | Institution: Ontario Tech University

1. Project Overview

This project extends Assignment 2 by training and comparing two supervised machine learning models to predict whether a financial customer is a high investment risk or not. The target variable is High Investment Risk with two possible values: No (not high risk) and Yes (high risk). This is a binary classification problem.

The two models compared are Logistic Regression and Support Vector Machine (SVM). Both models are evaluated using accuracy, precision, recall, and F1 score so the comparison is fair.

A key improvement over Assignment 2 is the use of `class_weight="balanced"` in both models. In Assignment 2, Logistic Regression was trained without this parameter and scored 0 percent on precision, recall and F1 because the severe class imbalance caused the model to predict No High Risk for almost every customer. In this assignment, class weighting is applied to both models to force them to pay more attention to the rare High Risk class during training.

2. Dataset

The Financial Customer Investment Risk Dataset contains 387 rows. After removing 7 duplicate rows, the cleaned dataset has 380 rows and 13 columns. The dataset is heavily imbalanced: 360 customers (94.7%) are labelled No High Risk and only 20 customers (5.3%) are labelled Yes High Risk.

Column	Type	Description
Customer_ID	Numerical	Unique identifier (not used as a feature)
Age	Numerical	Age of the customer in years
Employment_Status	Categorical	Full-time, Part-time, Retired, Student, Self-employed
Annual_Income	Numerical	Annual income in dollars
Investment_Experience_Years	Numerical	Years of investment experience

Portfolio_Value	Numerical	Total current investment value in dollars
Risk_Tolerance	Categorical	Low, Medium, or High
Number_of_Investment_Products	Numerical	Number of investment products held
Market_Volatility_Concern	Numerical	Score 1–5 for concern about market changes
Financial_Literacy_Score	Numerical	Score measuring financial knowledge
Advisor_Contacted	Categorical	Yes or No
Previous_Investment_Loss	Categorical	Yes or No

3. Methodology

The notebook follows 10 steps.

Step 1 : Understanding the Problem

Defines the business problem, identifies the target variable and all features, and explains why this is a supervised binary classification task. Introduces the two models to be compared and explains why class weighting is needed.

Step 2 : Data Inspection

Loads the raw dataset and checks the shape, column names, data types, missing values, duplicate rows, and class distribution.

Step 3 : Data Cleaning

Removes 7 duplicate rows. Fills missing numerical values with the column median and missing categorical values with the most common value. Saves the cleaned dataset.

Step 4 : Feature and Target Definition

Defines X (11 feature columns) and y (target column). Converts the target from text (No/Yes) to numbers (0/1).

Step 5 : Train and Test Split

Splits the data 80/20 with stratified sampling. Training set: 304 rows. Test set: 76 rows.

Step 6 : Preprocessing

Applies StandardScaler to numerical features and OneHotEncoder to categorical features using a ColumnTransformer. Fitted on training data only to prevent data leakage.

Step 7 : Training Logistic Regression

Trains LogisticRegression(max_iter=1000, random_state=42, class_weight="balanced").
The balanced weight is a direct improvement over Assignment 2 where its absence caused 0 percent recall.

Step 8 : Evaluating Logistic Regression

Evaluates the model using a confusion matrix, accuracy, precision, recall and F1 score.

Step 9 : Training and Evaluating SVM

Trains SVC(kernel="rbf", random_state=42, class_weight="balanced") and evaluates it using the same metrics for a fair comparison.

Step 10 : Model Comparison

Creates a side-by-side comparison table and a grouped bar chart of all four metrics for both models. Saves results to output/model_comparison_results.csv.

4. Results

4.1 Evaluation Metrics

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	92.11%	33.33%	50.00%	40.00%
SVM	92.11%	0.00%	0.00%	0.00%

4.2 Confusion Matrices

Logistic Regression:

	Predicted No High Risk	Predicted High Risk
Actual No High Risk	68 (True Negative)	4 (False Positive)
Actual High Risk	2 (False Negative)	2 (True Positive)

SVM:

	Predicted No High Risk	Predicted High Risk
Actual No High Risk	70 (True Negative)	2 (False Positive)
Actual High Risk	4 (False Negative)	0 (True Positive)

5. Interpretation

5.1 Which model performed better?

Logistic Regression performed better than SVM. Although both models achieved the same overall accuracy of 92.11 percent, Logistic Regression correctly identified 2 out of 4 actual High Risk customers (50 percent recall) while SVM identified none (0 percent recall). Logistic Regression also scored 33.33 percent precision and 40 percent F1, compared to 0 percent for SVM on both metrics.

5.2 Most important metric

Recall is the most important metric for this business problem. Missing a High Risk customer means that customer receives no intervention from an advisor and may face serious financial losses. Incorrectly flagging a safe customer only results in extra attention from an advisor which is not harmful. Therefore the goal is to maximise recall even at the cost of some false positives.

5.3 False positives and false negatives

A false positive means the model predicted High Risk but the customer is actually safe. Logistic Regression had 4 false positives and SVM had 2.

A false negative means the model predicted No High Risk but the customer is actually high risk. Logistic Regression had 2 false negatives and SVM had 4. Logistic Regression is preferred because fewer real high risk customers were missed.

6. Key Improvement Over Assignment 2

In Assignment 2, Logistic Regression was trained without `class_weight="balanced"` on the same dataset. The result was 93.42 percent accuracy but 0 percent precision, 0 percent recall and 0 percent F1 for the High Risk class. The model was predicting No for almost every customer.

In Assignment 3, `class_weight="balanced"` was added to both models as a direct response to this finding. For Logistic Regression this improved recall from 0 percent to 50 percent and F1 from 0 percent to 40 percent. SVM still scored 0 percent recall despite the same fix, which demonstrates that the RBF kernel struggles more with this level of class imbalance when only 16 High Risk examples are available in the training set.

7. Limitations and Recommendations

The most important limitation is the severe class imbalance. Only 20 out of 380 customers are labelled as High Risk. With only 16 High Risk examples in the training set

both models have very little information to learn from. Even with balanced class weighting, SVM still failed entirely and Logistic Regression only found half of the real high risk customers.

Recommended next steps to improve performance:

- Apply SMOTE (Synthetic Minority Oversampling Technique) to create synthetic High Risk examples and increase the minority class size before training.
- Try additional models such as Random Forest or Gradient Boosting which may handle imbalanced data more robustly than Logistic Regression or SVM.
- Collect more real High Risk customer data to give models a larger and more reliable set of minority examples to learn from.

Human judgment should always be part of the final decision. Even Logistic Regression missed 2 out of 4 High Risk customers. Financial advisors should use the model as a screening tool and apply their own knowledge when assessing borderline cases.